PRE PROCESSING PROJECT

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INTRODUCTION



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Education

Universitas Gadjah Mada
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Work Experience

Project Overhaul PLTU Sebalang
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PROJECT OVERVIEW

- Implementation of the Web-Based Internet of Things in Building Management Systems Aspects of Fire Protection and HVAC

  Tools : Arduino, Thingspeak

- Employee Performance Evaluation: Dashboard performa pegawai dan recomendasi untuk meningkatkan performa

  Tools : MS. Excel

MAIN PROJECT

- Problem Statement
  - Data Understanding
    - Data Preparation
      - Visualization
        - Recommendation

PROBLEM STATEMENT

- Mars Climate Orbiter NASA

Mars Climate Orbiter (MCO) diluncurkan pada 11 Desember 1998. Misinya adalah mempelajari atmosfer dan iklim Mars sekaligus menjadi relay komunikasi untuk Mars Polar Lander. Masalahnya terjadi pada data gaya kecil dari thruster. Software ground menghasilkan data dalam satuan pound-force second, tetapi Tim navigasi JPL menganggap angka tersebut sudah dalam Newton-second karena satuan itulah yang diwajibkan oleh dokumen interface. Hal ini mengakibatkan MCO mengorbit lebih rendah dari dari seharusnya dan mengakibatkan kerugian sekitar 125 juta USD.

Dari masalah itu, perlunya proses data cleaning yang tidak hanya memeriksa missing value, dan duplicate, tetapi juga memastikan konsistensi satuan, definisi, dan makna setiap variabel agar data dapat digunakan secara akurat dalam proses analisis dan pengambilan keputusan.

DATA UNDERSTANDING

- Data Set : ebay_womens_perfume.csv
- Range : 1000 rows & 10 columns
- Coloums : (brand, title, type), (price, priceWithCurrency), (available, availableText, sold), lastUpdated, itemLocation
- Missing Value : brand (0.1%), type (0.2%), available (13.1%), availableText (0.8%), sold (1.6%), lastUpdate (7.3%)
- Duplicated : 0.1 %

DATA PREPARATION

- Duplicates
- Missing Value
- Correct Inconsistencies
- Unnecessary columns
- Standardize Format
- Data types
- Outliers
- Feature Engineering

DUPLICATES

[8]:
```
df_raw[df_raw.duplicated(keep=False)]
```

[8]:

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation
344	FLOWER	12 Piece Woman's Perfume Lot: Assorted scents ...	Perfume	48.0	US $48.00	4.0	4 available / 1 sold	1.0	NaN	Webster, Florida, United States
662	FLOWER	12 Piece Woman's Perfume Lot: Assorted scents ...	Perfume	48.0	US $48.00	4.0	4 available / 1 sold	1.0	NaN	Webster, Florida, United States

[57]:
```
duplikat = df_raw.duplicated().sum() / len(df_raw) * 100   # % Duplikat
print ("Duplikat(%):",duplikat)
```

```
Duplikat(%): 0.1
```

Hapus duplikat

[58]:
```
df = df_raw.copy()
df = df.drop_duplicates()        #Data Frame baru untuk di cleaning
df[df.duplicated(keep=False)]
```

[58]:

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation

Hanya ada satu baris duplikat, untuk penanganannya adalah dengan dihapus

MISSING VALUE (BRAND, TITLE, DAN TYPE)

Mengisi mising kolom berdasarkan informasi "title"

[13]:
```
df[df["type"].isna()]
```

[13]:

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation
463	As Show	Carmina By Creed EDP 75ml 2.5 oz Spray For Wom...	NaN	128.88	US $128.88	10.0	10 available / 21 sold	21.0	May 24, 2024 00:32:16 PDT	New Jersey, United States, Hong Kong
856	NaN	Bright Crystal de Versace Eau de Toilette 3.0 ...	NaN	29.99	US $29.99	NaN	2 disponibles / 3 vendidos	NaN	23 may 2024 02:42:14 EST	Hamtramck, Michigan, Estados Unidos

[14]:
```
df.loc[463, ['brand', 'type']] = ['Creed', 'Eau de Parfum']
df.loc[856, ['brand', 'type']] = ['Versace', 'Eau de Toilette']
```

Karena kolom “title” berisi informasi tentang “brand” dan “type”, maka penanganan missing value pada “brand” dan “type” adalah dengan diisi sesuai informasi dari “title”

MISSING VALUE (AVAILABLETEXT)

Memisahkan missing value "availableText" menjadi dataframe baru

[24]:
```
df_discontinue = df[df["availableText"].isna()].copy()
df = df[df["availableText"].notna()].copy()
df_discontinue
```

[24]:

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation	price_c_clean
121	Ellis Brooklyn	NWB Ellis Brooklyn Miami Nectar Eau de Parfum ...	Eau de Parfum	33.00	US $33.00	NaN	NaN	NaN	NaN	Cranford, New Jersey, United States	33.00
602	Giardini Di Toscana	Bianco Latte EDP by Giardini Di Toscana 100 Ml...	Parfum	140.00	US $140.00	NaN	NaN	NaN	NaN	Houston, Texas, United States	140.00
732	Chanel	CHANEL Les Eaux De Chanel Paris Eau De Toilett...	Eau de Toilette	24.00	US $24.00	NaN	NaN	NaN	NaN	Los Angeles, California, United States	24.00
827	Ellis Brooklyn	NWB Ellis Brooklyn Biography Fragrance Discove...	Gift Set	17.00	US $17.00	NaN	NaN	NaN	NaN	Edison, New Jersey, United States	17.00
839	Lomani	SPICE ADDICT PRIVATE COLLECTION BY LOMANI EDP ...	Eau de Parfum	84.00	US $84.00	NaN	NaN	NaN	NaN	Hialeah, Florida, United States	84.00
880	Burberry	Burberry Goddess EDP Eau De Parfum Spray 3.3 o...	Eau de Parfum	95.00	US $95.00	NaN	NaN	NaN	NaN	West Hollywood, California, United States	95.00
927	Snif	Snif : Sweet Ash .07 FL. OZ. - Juniper, Fir Ba...	Parfum	14.95	US $14.95	NaN	NaN	NaN	NaN	Philadelphia, Pennsylvania, United States	14.95
983	Burberry	Burberry Her ELIXIR DE PARFUM EDP Intense-Delu...	Eau de Parfum	13.00	US $13.00	NaN	NaN	NaN	NaN	Chino, California, United States	13.00

"availableText" berisi informasi tentang available dan sold. Missing value pada kolom ini bisa menjadi indikasi bahwa produk-produk tersebut sudah tidak dijual lagi (discontinue). Sehingga untuk penanganan missing value kolom "availableText" Adalah dengan memisahkan dari data frame utama

MISSING VALUE (AVAILABLE)

Mengisi missing value jika pada "availableText" terdapat text "Last One"

[25]:
```
df.loc[
    df["available"].isna() &
    df["availableText"].str.contains("Last One", case=False, na=False),
    'available'
] = 1
```

Mengisi missing value jika pada "availableText" terdapat text "Out of Stock"

[26]:
```
df.loc[
    df["available"].isna() &
    df["availableText"].str.contains("Out of Stock", case=False, na=False),
    "available"
] = 0
```

```
# input manual
df.loc[990, 'available'] = 32
```

Mengisi missing value jika pada "availableText" terdapat jumlah available-nya

[27]:
```
mask = df["available"].isna()

# untuk "n available"
available_extract = (
    df.loc[mask, "availableText"]
    .str.extract(r'(\d+)\s+available', expand=False)
)

df.loc[mask, "available"] = (
    df.loc[mask, 'available']
    .fillna(pd.to_numeric(available_extract, errors='coerce'))
)

# untuk lot dan jumlah per-lot
lots = df.loc[mask, 'availableText'].str.extract(
    r'(\d+)\s+lots?\s+available',
    expand=False
)

items_per_lot = df.loc[mask, 'availableText'].str.extract(
    r'(\d+)\s+items?\s+per\s+lot',
    expand=False
)

total_items = (
    pd.to_numeric(lots, errors='coerce') *
    pd.to_numeric(items_per_lot, errors='coerce')
)

df.loc[mask, 'available'] = (
    df.loc[mask, 'available']
    .fillna(total_items)
)

# untuk bahasa spanyol
mask_spanish = (
    df['available'].isna() &
    df['availableText'].str.contains(
        r'\d+\s+disponibles?',
        case=False,
        na=False
    )
)

df.loc[mask_spanish, 'available'] = (
    df.loc[mask_spanish, 'availableText']
    .str.extract(r'(\d+)\s+disponibles?', expand=False)
    .astype(int)
```

MISSING VALUE (AVAILABLE)

Mengisi missing value dengan median untuk Limited quantity

[28]:
```
mask_limited = (
    df['available'].isna() &
    df['availableText'].str.contains(
        'Limited quantity available',
        case=False,
        na=False
    )
)

df.loc[mask_limited, 'available'] = df['available'].median()
```

Untuk missing value yang tidak terdapat informasi dapat “availableText”, maka penanganannya adalah dengan imputasi nilai median. Hal ini diharapkan tidak menggangu bentuk distribusi data

MISSING VALUE (SOLD)

Mengatasi missing value untuk kolom "sold"

[29]:
```
df[df["sold"].isna()]
```

[29]:

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation	price_c_clean
4	Khadlaj	Shiyaaka for Men EDP Spray 100ML (3.4 FL.OZ) B...	Eau de Parfum	29.99	US $29.99/ea	10.0	More than 10 available	NaN	NaN	Little Ferry, New Jersey, United States	29.99
114	Prada	Prada Paradoxe EDP Mini Bottle 7ml/.23oz	Eau de Parfum	19.00	US $19.00	10.0	More than 10 available	NaN	May 23, 2024 06:27:16 PDT	Bowling Green, Kentucky, United States	19.00
117	Ouai	OUAI Hair & Body Mist In ST. BARTS 3.3oz / 97m...	Fragrance Mist	28.00	US $28.00/ea	2.0	2 available	NaN	NaN	Beverly Hills, California, United States	28.00
432	Maison Alhambra	DELILAH PERFUME EDP 3.4 OZ BY MAISON ALHAMBRA ...	Eau de Parfum	23.99	US $23.99	10.0	More than 10 available	NaN	NaN	New York, New York, United States	23.99
496	Victoria'S Secret	Victoria's Secret Bombshell Sundrenched Eau De...	Eau de Parfum	59.99	US $59.99/ea	4.0	4 available	NaN	NaN	Colorado Springs, Colorado, United States	59.99
551	Jean Couturier	CORIANDRE by JEAN COUTURIER for Women 30ml-1oz...	Eau de Toilette	21.95	US $21.95/ea	5.0	5 available	NaN	NaN	Los Angeles, California, United States	21.95
617	Dossier	Dossier MUSKY MUSK Eau de Parfum 1.7 Fl oz / 5...	Eau de Parfum	20.00	US $20.00/ea	10.0	More than 10 available	NaN	NaN	Los Angeles, California, United States	20.00
856	Versace	Bright Crystal de Versace Eau de Toilette 3.0 ...	Eau de Toilette	29.99	US $29.99	2.0	2 disponibles / 3 vendidos	NaN	23 may 2024 02:42:14 EST	Hamtramck, Michigan, Estados Unidos	29.99

[30]:
```
# input manual karna tertera pada "availableText"
df.loc[856, 'sold'] = 3

# mengisi dengan median
df["sold"] = df['sold'].fillna(df['sold'].median())
```

Sama seperti kolom “available”, Untuk missing value yang tidak terdapat informasi dapat “availableText”, maka penanganannya adalah dengan imputasi nilai median. Hal ini diharapkan tidak menggangu bentuk distribusi data

CORRECT INCONSISTENCIES

Hapus spasi

[12]:
```
df[["brand","title","type"]]=df[["brand","title","type"]].apply(
    lambda x : x.str.strip().str.replace(r'\s+', ' ', regex=True)
)
df
```

[12]:

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation
0	Carolina Herrera	Good Girl by Carolina Herrera 2.7 oz Eau De Pa...	Eau de Parfum	43.99	US $43.99/ea	2.0	2 available / 393 sold	393.0	May 23, 2024 10:43:50 PDT	Thomasville, Alabama, United States
1	As Shown	Parfums de Marly Delina La Rosee Eau de Parfum...	Eau de Parfum	79.99	US $79.99	5.0	5 available / 40 sold	40.0	May 24, 2024 00:15:48 PDT	New Jersey, Hong Kong
2	PRADA	PRADA Paradoxe by Prada EDP 3.0oz/90ml Spray P...	Eau de Parfum	59.99	US $59.99	10.0	More than 10 available / 35 sold	35.0	May 14, 2024 20:54:25 PDT	Orange, New Jersey, United States
3	As Show	J'adore Parfum D'eau by Christian 3.4 oz EDP F...	Eau de Parfum	59.99	US $59.99/ea	10.0	More than 10 available / 9 sold	9.0	May 23, 2024 01:23:05 PDT	USA, New Jersey, Hong Kong
4	Khadlaj	Shiyaaka for Men EDP Spray 100ML (3.4 FL.OZ) B...	Eau de Parfum	29.99	US $29.99/ea	10.0	More than 10 available	NaN	NaN	Little Ferry, New Jersey, United States
...	...	...	...	...	...	...	...	...	...	...
995	Avon	Avon Far Away Infinity Eau de Parfum 1.7 fl. o...	Eau de Parfum	13.89	US $13.89	10.0	More than 10 available / 157 sold	157.0	May 16, 2024 22:35:29 PDT	West Palm Beach, Florida, United States
996	Mancera	Roses Greedy by Mancera perfume for unisex EDP...	Eau de Parfum	57.85	US $57.85/ea	33.0	33 available / 58 sold	58.0	May 24, 2024 08:03:11 PDT	Dallas, Texas, United States
997	Unbranded	Sweet Tooth Eau de Parfum, Perfume for Women, ...	1	30.96	US $30.96	2.0	2 available / 3 sold	3.0	May 17, 2024 23:16:41 PDT	New York, New York, United States
998	Juliette Has A Gun	MMMM BY Juliette Has A Gun perfume for her EDP...	Eau de Perfume	53.99	US $53.99/ea	3.0	3 available / 117 sold	117.0	May 13, 2024 22:19:34 PDT	Dallas, Texas, United States
999	Paris Hilton	PARIS HILTON ELECTRIFY for Women Cologne 3.4 o...	Eau de Parfum	14.99	US $14.99/ea	4.0	4 available / 51 sold	51.0	May 22, 2024 05:44:45 PDT	TX, United States

CORRECT INCONSISTENCIES (BRAND)

Membuat nama "brand" lebih konsisten

[15]:
```
pd.set_option('display.max_rows', None)
df["brand"] = df["brand"].str.title()
df["brand"].value_counts().sort_index(ascending=True)
```

[15]:
```
brand
Acqua Di Parma              2
Adam Levine                 1
Aerin                       2
Aeropostale                 1
Afnan                       3
Ainash Perfums              1
Al Hambra                   1
Al Haramain                 2
Al Rehab                    2
Al Wataniah                 2
Al-Rehab                    1
Alfred Sung                 5
Alt Fragrances              2
Anna Sui                    2
Aquolina                    2
Ariana Grande              25
Armaf                       8
```

[17]:
```
df[df["brand"] == "Tf"]     #Filter untuk membantu pembuatan mapping
```

[16]:
```
brand_mapping = {
    'Dolce&Gabbana': 'Dolce & Gabbana',
    'Dolce Gabbana': 'Dolce & Gabbana',

    'Ysl': 'Yves Saint Laurent',

    'Giorgi^O Armani': 'Giorgio Armani',
    'Giorgio Arm.Ani': 'Giorgio Armani',
    'Giorgio² Armani': 'Giorgio Armani',

    'Lancome': 'Lancôme',

    'Estee Lauder': 'Estée Lauder',

    'Chloe': 'Chloé',

    'Parfums Gres': 'Parfums Grès',

    'Al-Rehab': 'Al Rehab',

    'J Lo': 'Jennifer Lopez',
    'Jlo': 'Jennifer Lopez',

    'Lattafa Perfumes': 'Lattafa',

    'Kate Spade New York': 'Kate Spade',

    'Maison Martin Margiela': 'Maison Margiela',

    'Elizabeth And James Nirvana': 'Elizabeth & James',

    'Gianni Versace': 'Versace',

    'Benetton': 'United Colors Of Benetton'
}
df["brand"] = df["brand"].replace(brand_mapping)
df["brand"].value_counts().sort_index(ascending=True)
```

Mapping ada dictionary yang berisi penulisan brand-brand typo yang kemudian diperbaiki penulisannya

CORRECT INCONSISTENCIES (TYPE)

Membuat nama "type" lebih konsisten

[18]:
```
df["type"] = (
    df["type"].str.lower()
)
df["type"].value_counts(dropna=False).sort_index(ascending=True)
```

[18]:
```
type
/                                          1
1                                          1
3 pc                                       1
asst                                       1
bath oil                                   1
beauty                                     1
body lotion                                1
body mist                                  1
body powder                                1
body spray                                 1
cologne                                    3
cologne spray                              1
cream                                      1
deodorant                                  1
deodorant stick                            1
discovery set                              1
does not apply                             2
eau de cologne                            38
eau de parfum                            579
eau de parfum 2 pcs set                    1
eau de parfum 3 pcs set                    1
eau de parfum spray (unisex tester) 3.4 oz 1
eau de parfum supreme                      1
eau de parfum, eau de parfume              1
eau de parfum, spray                       1
eau de parfum/perfume                      1
eau de parfume                             2
eau de perfume                            23
eau de toilette                          192
eau de toilette spray 3.4 oz               1
eau de toilette, spray                     6
edp                                        1
edp and parfum                             1
edt                                        1
elixir de parfum                           1
esprit de parfum                           2
extract parfum                             1
extrait de parfum                          3
fine fragrance mist                        1
```

[19]:
```
type_mapping = {
    'eau de parfum': 'Eau de Parfum',
    'eau de perfume': 'Eau de Parfum',
    'eau de parfume': 'Eau de Parfum',
    'edp': 'Eau de Parfum',
    "l'eau de parfum": 'Eau de Parfum',
    'eau de parfum, spray': 'Eau de Parfum',
    'eau de parfum, eau de parfume': 'Eau de Parfum',
    'eau de parfum spray (unisex tester) 3.4 oz': 'Eau de Parfum',
    'eau de parfum supreme': 'Eau de Parfum',
    'eau de parfum 2 pcs set': 'Eau de Parfum',
    'eau de parfum 3 pcs set': 'Eau de Parfum',
    'eau de parfum/perfume': 'Eau de Parfum',
    'perfume, eau de parfum': 'Eau de Parfum',
    'edp and parfum': 'Eau de Parfum',
    '/': 'Eau de Parfum',
    '1': 'Eau de Parfum',


    'eau de toilette': 'Eau de Toilette',
    'edt': 'Eau de Toilette',
    'eau de toilette, spray': 'Eau de Toilette',
    'toilette spray': 'Eau de Toilette',
    'eau de toilette spray 3.4 oz': 'Eau de Toilette',

    'eau de cologne': 'Eau de Cologne',
    'cologne': 'Eau de Cologne',
    'cologne spray': 'Eau de Cologne',

    'parfum': 'Parfum',
    'perfume': 'Parfum',
    'extrait de parfum': 'Parfum',
    'extract parfum': 'Parfum',
    'esprit de parfum': 'Parfum',
    'elixir de parfum': 'Parfum',

    'fragrance mist': 'Fragrance Mist',
    'fine fragrance mist': 'Fragrance Mist',
    'fragrance body mist': 'Fragrance Mist',
    'perfume fragrance mist': 'Fragrance Mist',
    'body mist': 'Fragrance Mist',
    'mist': 'Fragrance Mist',

    'perfume oil': 'Perfume Oil',
    'oil perfume': 'Perfume Oil',
    'scented oils': 'Perfume Oil',

    'body spray': 'Body Spray',
    'spray': 'Body Spray',

    'lotion': 'Body Lotion',
    'body lotion': 'Body Lotion',
    'skin_moisturizer': 'Body Lotion',
    'sensuous body moisturizer': 'Body Lotion',

    'deodorant': 'Deodorant',
    'deodorant stick': 'Deodorant',

    'gift sets': 'Gift Set',
    'perfume gift sets': 'Gift Set',
    'discovery set': 'Gift Set'
}
```

[20]:
```
df["type"] = df["type"].replace(type_mapping)
df["type"].value_counts(dropna=False).sort_index(ascending=True)
```

[20]:
```
type
3 pc                                       1
Body Lotion                                8
Body Spray                                 6
Deodorant                                  2
Eau de Cologne                            42
Eau de Parfum                            619
Eau de Toilette                          201
Fragrance Mist                            16
Gift Set                                   6
Parfum                                    65
Perfume Oil                                5
asst                                       1
bath oil                                   1
beauty                                     1
body powder                                1
cream                                      1
does not apply                             2
fragrances                                14
hair perfume                               1
parfum, lotion, gloss and blush            1
pink sugar                                 1
shimmer                                    1
shimmering body oil                        1
solid perfume stick                        1
~ body firm advanced body repair treatment ~  1
Name: count, dtype: int64
```

[21]:
```
df[df["type"] == "3 pc"]     #Filter untuk membantu pembuatan mapping
```

[21]:

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation
74	Elizabeth Arden	Elizabeth Arden White Tea Fragrance Collection...	3 pc	19.45	US $19.45	50.0	50 available / 2 sold	2.0	May 24, 2024 07:12:19 PDT	Ronkonkoma, New York, United States

UNNECESSARY COLUMNS

Mengambil value angka pada colom "priceWithCurrency"

[22]:
```
df["price_c_clean"] = (
    df["priceWithCurrency"]
    .str.extract(r'(\d+(?:\.\d+)?)')[0]
    .astype(float)
)
df[["price","price_c_clean"]]
```

[22]:

	price	price_c_clean
0	43.99	43.99
1	79.99	79.99
2	59.99	59.99
3	59.99	59.99
4	29.99	29.99
5	51.99	51.99
6	58.99	58.99
7	52.79	52.79
8	29.99	29.99

Membandingkan dengan kolom "price"

[23]:
```
df[df["price"] != df["price_c_clean"]]
```

[23]:

brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation	price_c_clean

kedua kolom memiliki value yang sama sehingga salah satu kolom bisa diabaikan

Ketika dibandingkan value antara kolom “price” dan “priceWithCurrency”, tidak ada perbedaan antara dua kolom tersebut. Sehingga salah satu kolom dapat diabaikan

STANDARDIZE FORMAT

Menghapus teks pada kolom ¶

[31]:
```
df["lastUpdated"] = df["lastUpdated"].str.strip()
df["lastUpdated"] = (
    df['lastUpdated']
    .str.replace(r'\s+(PDT|PST|EST)$', '', regex=True)
)
```

Mengubah tipe data ke datetime

[32]:
```
df["lastUpdated"] = pd.to_datetime(
    df["lastUpdated"],
    format = "mixed",
    errors = "coerce"
)
```

Memisahkan tanggal dan waktu ¶

[33]:
```
df["Date"] = df["lastUpdated"].dt.date
df["Time"] = df["lastUpdated"].dt.time
```

DATA TYPES

```
[34]: df.info()
```

```
<class 'pandas.DataFrame'>
Index: 991 entries, 0 to 999
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   brand              991 non-null    str
 1   title              991 non-null    str
 2   type               991 non-null    str
 3   price              991 non-null    float64
 4   priceWithCurrency  991 non-null    str
 5   available          991 non-null    float64
 6   availableText      991 non-null    str
 7   sold               991 non-null    float64
 8   lastUpdated        927 non-null    datetime64[us]
 9   itemLocation       991 non-null    str
 10  price_c_clean      991 non-null    float64
 11  Date               927 non-null    object
 12  Time               927 non-null    object
dtypes: datetime64[us](1), float64(4), object(2), str(6)
memory usage: 298.8+ KB
```

Memperbaiki tipe data

```
[35]: df[["available","sold"]] = df[["available","sold"]].astype(int)
df["Date"] = pd.to_datetime(df["Date"])
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 991 entries, 0 to 999
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   brand              991 non-null    str
 1   title              991 non-null    str
 2   type               991 non-null    str
 3   price              991 non-null    float64
 4   priceWithCurrency  991 non-null    str
 5   available          991 non-null    int64
 6   availableText      991 non-null    str
 7   sold               991 non-null    int64
 8   lastUpdated        927 non-null    datetime64[us]
 9   itemLocation       991 non-null    str
 10  price_c_clean      991 non-null    float64
 11  Date               927 non-null    datetime64[s]
 12  Time               927 non-null    object
dtypes: datetime64[s](1), datetime64[us](1), float64(2), int64(2), object(1), str(6)
memory usage: 298.8+ KB
```

OUTLIERS (TYPE)

```
[37]: type_counts = df["type"].value_counts()

      Q1 = type_counts.quantile(0.25)
      Q3 = type_counts.quantile(0.75)

      IQR = Q3 - Q1

      lower = Q1 - 1.5 * IQR
      upper = Q3 + 1.5 * IQR

      type_outlier = type_counts[
          (type_counts < lower) |
          (type_counts > upper)
      ]

      print("Q1:", Q1)
      print("Q3:", Q3)
      print("IQR:", IQR)
      print("Lower Bound:", lower)
      print("Upper Bound:", upper)
      type_outlier
```

```
Q1: 1.0
Q3: 8.0
IQR: 7.0
Lower Bound: -9.5
Upper Bound: 18.5
```

```
[37]: type
      Eau de Parfum      615
      Eau de Toilette    200
      Parfum              63
      Eau de Cologne      42
      Name: count, dtype: int64
```

Ketika dilakukan pengecekan outlier dengan metode IQR, “type” dengan jumlah banyak dianggap sebagai outlier. Sehingga, penetapan outlier dengan IQR anggap kurang relevan dengan kolom ini

OUTLIERS (TYPE)

```
[38]: df["type"].value_counts()
```

```
[38]: type
Eau de Parfum                                   615
Eau de Toilette                                 200
Parfum                                           63
Eau de Cologne                                   42
Fragrance Mist                                   16
fragrances                                       14
Body Lotion                                       8
Body Spray                                        6
Gift Set                                          5
Perfume Oil                                       5
Deodorant                                         2
does not apply                                    2
~ body firm advanced body repair treatment ~      1
3 pc                                              1
hair perfume                                      1
cream                                             1
shimmer                                           1
parfum, lotion, gloss and blush                   1
bath oil                                          1
asst                                              1
beauty                                            1
shimmering body oil                               1
body powder                                       1
solid perfume stick                               1
pink sugar                                        1
Name: count, dtype: int64
```

```
[39]: type_counts = df["type"].value_counts()
      df["type"] = df["type"].apply(
          lambda x: x if type_counts[x] > 2 else "Others")

      df["type"].value_counts()
```

```
[39]: type
Eau de Parfum      615
Eau de Toilette    200
Parfum              63
Eau de Cologne      42
Others              17
Fragrance Mist      16
fragrances          14
Body Lotion          8
Body Spray           6
Gift Set             5
Perfume Oil          5
Name: count, dtype: int64
```

OUTLIERS (PRICE)

Outlier kolom price

```
[40]: Q1 = df["price"].quantile(0.25)
      Q3 = df["price"].quantile(0.75)

      IQR = Q3 - Q1

      lower = Q1 - 1.5 * IQR
      upper = Q3 + 1.5 * IQR

      price_outlier = df[
          (df["price"] < lower) |
          (df["price"] > upper)
      ]

      print("Q1:", Q1)
      print("Q3:", Q3)
      print("IQR:", IQR)
      print("Lower Bound:", lower)
      print("Upper Bound:", upper)
      price_outlier["price"].count()
```

```
Q1: 20.75
Q3: 49.99
IQR: 29.240000000000002
Lower Bound: -23.11
Upper Bound: 93.85

[40]: np.int64(50)
```

Melihat harga-harga parfum yang sangat beragam, maka untuk nilai outlier “price” tidak dilakukan penanganan (dibiarkan)

OUTLIERS (AVAILABLE)

Outlier kolom available

[41]:
```
Q1 = df["available"].quantile(0.25)
Q3 = df["available"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

available_outlier = df[
    (df["available"] < lower) |
    (df["available"] > upper)
]

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower)
print("Upper Bound:", upper)
available_outlier["available"].count()
```

```
Q1: 5.0
Q3: 10.0
IQR: 5.0
Lower Bound: -2.5
Upper Bound: 17.5
```

[41]:
```
np.int64(120)
```

Karna sangat mungkin jumlah stok bernilai besar, maka untuk nilai outlier “available” tidak dilakukan penanganan (dibiarkan)

OUTLIERS (SOLD)

Outlier kolom sold

[42]:
```
Q1 = df["sold"].quantile(0.25)
Q3 = df["sold"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

sold_outlier = df[
    (df["sold"] < lower) |
    (df["sold"] > upper)
]

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower)
print("Upper Bound:", upper)
sold_outlier["sold"].count()
```

```
Q1: 15.0
Q3: 258.0
IQR: 243.0
Lower Bound: -349.5
Upper Bound: 622.5
```

[42]:
```
np.int64(148)
```

Karna sangat mungkin jumlah terjual bernilai besar, maka untuk nilai outlier “sold” tidak dilakukan penanganan (dibiarkan)

FEATURE ENGINEERING

```
[47]: df_clean["Potential_Revenue"] = df_clean["Price"]*df_clean["Available"]
      df_clean["Estimated_Total_Sales"] = df_clean["Price"]*df_clean["Sold"]
      df_clean
```

[47]:

	Brand	Title	Type	Price	Available	Sold	Item_Location	Last_Update(Date)	Potential_Revenue	Estimated_Total_Sales
0	Carolina Herrera	Good Girl by Carolina Herrera 2.7 oz Eau De Pa...	Eau de Parfum	43.99	2	393	Thomasville, Alabama, United States	2024-05-23	87.98	17288.07
1	As Shown	Parfums de Marly Delina La Rosee Eau de Parfum...	Eau de Parfum	79.99	5	40	New Jersey, Hong Kong	2024-05-24	399.95	3199.60
2	Prada	PRADA Paradoxe by Prada EDP 3.0oz/90ml Spray P...	Eau de Parfum	59.99	10	35	Orange, New Jersey, United States	2024-05-14	599.90	2099.65
3	As Show	J'adore Parfum D'eau by Christian 3.4 oz EDP F...	Eau de Parfum	59.99	10	9	USA, New Jersey, Hong Kong	2024-05-23	599.90	539.91

```
[48]: print("Potential Revenue:",df_clean["Potential_Revenue"].sum())
      print("Estimated Total Sales:",df_clean["Estimated_Total_Sales"].sum().round(2))
```

```
Potential Revenue: 571049.99
Estimated Total Sales: 13898684.79
```

Feature Engineering dilakukan untuk melihat potensi pendapatan dan juga perkiraan penjualan. Terlihat bahwa terdapat potensi pendapatan sekitar 571.049 USD

VISUALIZATION

[46]:
```
df_clean = df[["brand","title","type","price","available","sold","itemLocation","Date"]].copy()
df_clean.reset_index(drop=True)
df_clean = df_clean.rename(columns={
    "brand" : "Brand",
    "title" : "Title",
    "type" : "Type",
    "price" : "Price",
    "available" : "Available",
    "sold" : "Sold",
    "itemLocation" : "Item_Location",
    "Date" : "Last_Update(Date)"
})
df_clean
```

[46]:

	Brand	Title	Type	Price	Available	Sold	Item_Location	Last_Update(Date)
0	Carolina Herrera	Good Girl by Carolina Herrera 2.7 oz Eau De Pa...	Eau de Parfum	43.99	2	393	Thomasville, Alabama, United States	2024-05-23
1	As Shown	Parfums de Marly Delina La Rosee Eau de Parfum...	Eau de Parfum	79.99	5	40	New Jersey, Hong Kong	2024-05-24
2	Prada	PRADA Paradoxe by Prada EDP 3.0oz/90ml Spray P...	Eau de Parfum	59.99	10	35	Orange, New Jersey, United States	2024-05-14
3	As Show	J'adore Parfum D'eau by Christian 3.4 oz EDP F...	Eau de Parfum	59.99	10	9	USA, New Jersey, Hong Kong	2024-05-23
4	Khadlaj	Shiyaaka for Men EDP Spray 100ML (3.4 FL.OZ) B...	Eau de Parfum	29.99	10	52	Little Ferry, New Jersey, United States	NaT
5	Viktor & Rolf	Flowerbomb by Viktor & Rolf 3.4 oz 100ML Eau D...	Eau de Parfum	51.99	8	184	Warren, Michigan, United States	2024-05-23

VISUALIZATION

[44]:
```
df.describe()
```

[44]:

	price	available	sold	lastUpdated	price_c_clean	Date
count	991.000000	991.000000	991.000000	927	991.000000	927
mean	39.782069	19.397578	494.177598	2024-05-15 14:00:38.570657	39.782069	2024-05-15 02:46:12
min	1.990000	0.000000	1.000000	2023-07-22 17:43:51	1.990000	2023-07-22 00:00:00
25%	20.750000	5.000000	15.000000	2024-05-16 12:20:20	20.750000	2024-05-16 00:00:00
50%	32.990000	10.000000	52.000000	2024-05-22 07:44:52	32.990000	2024-05-22 00:00:00
75%	49.990000	10.000000	258.000000	2024-05-23 21:05:20	49.990000	2024-05-23 00:00:00
max	299.990000	557.000000	17854.000000	2024-05-24 11:16:44	299.990000	2024-05-24 00:00:00
std	28.904156	48.549104	1368.157831	NaN	28.904156	NaN

[45]:
```
df[df.duplicated()]
```

[45]:

brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation	price_c_clean	Date	Time

[43]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 991 entries, 0 to 999
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   brand              991 non-null    str
 1   title              991 non-null    str
 2   type               991 non-null    str
 3   price              991 non-null    float64
 4   priceWithCurrency  991 non-null    str
 5   available          991 non-null    int64
 6   availableText      991 non-null    str
 7   sold               991 non-null    int64
 8   lastUpdated        927 non-null    datetime64[us]
 9   itemLocation       991 non-null    str
 10  price_c_clean      991 non-null    float64
 11  Date               927 non-null    datetime64[s]
 12  Time               927 non-null    object
dtypes: datetime64[s](1), datetime64[us](1), float64(2), int64(2), object(1), str(6)
memory usage: 298.5+ KB
```

Dari informasi data yang sudah dicleaning, Perusahaan bisa menjadikan rata-rata “price” sebagai patokan awal untuk membuat promo menarik sehingga potensi pendapatan sekitar 571.049 USD bisa dimaksimalkan

RECOMMENDATION

- Perusahaan perlu memvalidasi beberapa “title” parfum yang terindikasi discontinue kepada pihak brand
- Perudahaan memiliki potensi pendapatan sekitar 571.049 USD sehingga diperlukan strategi penjualan seperti promo di jam tertentu untuk memaksimalkan potensi pendapatan